

MEMO — Milestone 2: GJR-GARCH Volatility Forecasting & VaR Backtesting

Swiss Market Forecasting Project | UBSG (separate) vs. NESN/NOVN/ROP (pooled) |
Backtest window: 2021-01-04 to present (1,397 trading days)

RECOMMENDATION

Deploy GJR-GARCH(1,1), Student-t, UBSG fit separately (LR-test supported, standalone post-merger $p < 10^{-8}$) from pooled NESN/NOVN/ROP. 1-day VaR backtest is clean at 95% for all four names; UBSG's 99% VaR shows a borderline, uncorrected-significant undercoverage signal (does not survive multiple-testing correction) traced to a genuine skew/tail-mismatch in UBSG's innovations, not a fitting artifact — both persistence and event-clustering hypotheses were tested and ruled out. For 10-day stress: report FHS as primary, MC as model-diagnostic cross-check — MC's parametric assumption is measurably too thin-tailed. Recommend proceeding to Milestone 3 with skewed-t innovation testing for UBSG as the top backlog item.

KEY FINDINGS

Item	Finding	Evidence
Separate vs. pooled (UBSG)	LR rejects full pooling ($p \approx 0$, full sample). Standalone post-merger-only test also significant ($p = 9.7 \times 10^{-8}$, $n = 780$) despite short window — pre-merger-only test also rejects.	Sec. 1–2
Fitting bug #1 (caught & fixed)	First LR test returned an impossible negative statistic — local-optimum failure, not a real result. Fixed: multi-start + warm-start from restricted solution.	Sec. 1
Fitting bug #2 (caught & fixed)	11/23 UBSG walk-forward refits showed omega pinned at its lower bound (near-IGARCH degeneracy). Fixed via variance-targeting. Re-run: UBSG 95% VaR clean.	Sec. 4–5
1-day VaR backtest	95% clean for all 4 names. UBSG 99%: uncorrected Kupiec $p = 0.014$, does NOT survive Bonferroni or BH-FDR correction across the 8 tests run. Christoffersen conditional coverage also non-significant.	Sec. 3, 6
Persistence hypothesis	RULED OUT: near-unit persistence (0.997–0.999) is present in ALL 23 refits, not just some — no cross-quarter variation to explain which quarters had violations.	Sec. 7a
Event-clustering hypothesis (SVB/CS crisis)	BUILT AND KILLED: refit 8 (Mar 2023, SVB/CS collapse) had 3 violations, the most of any quarter. Simulated null: $P(\max \text{violations across } 23 \text{ quarters} \geq 3) = 0.22$ — not significant.	Sec. 7b
Skew / tail-mismatch (what survived)	UBSG has the most negative skew (−0.79) and largest one-sided loss-tail gap (−25.9%) of the 4 names on true out-of-sample residuals — the specific channel that explains the 99% VaR undercoverage.	Sec. 7 (table)
10-day stress: MC vs. FHS	FHS more extreme than MC in 8/8 name $\times$ level comparisons (5.9%–11.7%). Traced to the same thin-tail/skew issue as the 1-day finding — same underlying cause.	Sec. 8

LIMITATIONS

UBSG separate-treatment and skew findings both partly rest on a ~ 2 -year post-merger window (780 obs) — current best evidence, not permanent. 99% VaR backtest power remains limited (~ 14 expected violations) despite the extended window. Event-clustering tested with a simple max-of-23 null, not a full event-study design — legitimate lower-priority follow-up, not fully closed. MC/FHS comparison uses a single full-sample fit (Sec. 8 scope), not a walk-forward version.

NEXT STEP (MILESTONE 3)

Test skewed-t / nonparametric innovation distributions for UBSG — motivated by the walk-forward-verified skew/tail-mismatch that survived every check run against competing explanations (persistence: ruled out; event-clustering: built and killed). Event-clustering retained as a lower-priority thread if a proper event-study design becomes worth building. ML challenger model (LSTM / gradient boosting on realized vol) as previously scoped.

Full detail, code, and the complete chronological record (including both fitting bugs and the killed event-clustering narrative): `notebooks/02_garch_var_backtest.ipynb`.